Symplectic Geometry by Xiaobo Liu

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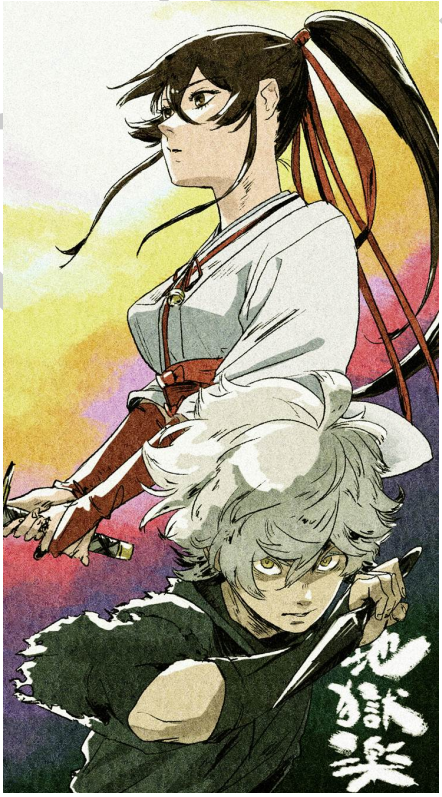

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1 March 03, 2026: Introduction

References

(1) Da Silva, "Lectures on Symplectic Geometry";

(2) McDuff-Salamon, "Introduction to Symplectic Topology";

(3) McDuff-Salamon, "J-Holomorphic Curves and Symplectic Topology".

1.1 Introduction

Comparison with (Pseudo) Riemannian Geometry

(Pseudo) Riemannian Geometry	**Symplectic Geometry**
Metric g on tangent bundle TM	Non-degenerate closed 2-form ω on M
$\forall p \in M, g_p : T_pM \times T_pM \to \mathbb{R}$ **symmetric** bilinear non-degenerate	$\forall p \in M, \omega_p : T_pM \times T_pM \to \mathbb{R}$ **skew-symmetric** bilinear non-degenerate
g is a symmetric non-degenerate 2-tensor	ω is a **closed** skew-symmetric non-degenerate 2-form
g always exists on any smooth manifold M	ω may not exist on M
$\dim M$ can be any positive integer	$\dim M$ must be even
Have small symmetry group (isometry group)	If $\dim M > 0$, the symmetry group is infinite-dimensional
Have local invariants (curvature)	No local invariants
In general, no almost complex structure	Always admit an almost complex structure
Applications in physics: general relativity	Applications in physics: classical mechanics, string theory, etc.

Definition 1.1. Let $V \cong \mathbb{R}^n$ be a n-dimensional real vector space. A *symplectic form* on V is a skew-symmetric bilinear non-degenerate form $B : V \times V \to \mathbb{R}$. The pair (V, B) is called a *symplectic vector space*.

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Lemma 1.2. Let V be a finite-dimensional real vector space and $B : V \times V \to \mathbb{R}$ be a skew-symmetric bilinear form. TFAE:

(1) the linear map $V \to V^*, v \mapsto B(v, \cdot)$ is an isomorphism;

(2) choose a basis $\{e_1, \dots, e_n\}$ of V, then the matrix $B_{ij} := B(e_i, e_j)$ is invertible;

(3) $n = 2m$, and there exists a basis $\{e_1, \dots, e_{2m}\}$ of V such that $B = e_1^* \wedge e_2^* + \cdots + e_{2m-1}^* \wedge e_{2m}^*$;

(4) $n = 2m$, regard $B \in V^* \wedge V^*$, then $B^m = B \wedge \cdots \wedge B \in \bigwedge^n V^* \cong \mathbb{R}$ is nonzero.

If any of the above conditions holds, we say B is non-degenerate.

Proof. Homework. □

Corollary 1.3. Any two symplectic vector spaces of the same dimension are isomorphic.

Definition 1.4. A *symplectic form* on a smooth manifold M is a closed non-degenerate 2-form $\omega \in \Omega^2(M)$. Here, ω is non-degenerate means that for any $p \in M$, the bilinear form $\omega_p : T_pM \times T_pM \to \mathbb{R}$ is non-degenerate. The pair (M, ω) is called a *symplectic manifold*.

Corollary 1.5. Let (M, ω) be a symplectic manifold. Then

(a) we have an isomorphism $TM \cong T^*M$ of vector bundles given by $v \mapsto \omega(v, \cdot)$;

(b) $\dim M = 2m$ is even, and $\omega^m := \omega \wedge \cdots \wedge \omega \in \Omega^{2m}(M)$ is nowhere vanishing, hence defines a volume form on M. In particular, M is orientable.

Recall the *de Rham* cohomology

$$H^k_{\mathrm{dR}}(M) := \frac{\ker(d : \Omega^k(M) \to \Omega^{k+1}(M))}{\mathrm{im}(d : \Omega^{k-1}(M) \to \Omega^k(M))}.$$

The symplectic form ω defines a cohomology class $[\omega] \in H^2_{\mathrm{dR}}(M)$.

2 March 05: first properties and more examples

Recall: let ω be a symplectic form on M. Since it is closed, it defines a de Rham cohomology class $[\omega] \in H^2_{dR}(M, \mathbb{R})$.

Theorem 2.1. If M is compact with a symplectic form ω, then $[\omega] \neq 0 \in H^2_{dR}(M, \mathbb{R})$.

Proof. Suppose the contrary that $[\omega] = 0$. Then there exists a 1-form θ such that $\omega = d\theta$. Let $n = 2m = \dim M$. Consider the top form $\omega^m = d(\theta \wedge \omega^{m-1})$. Integrating over M, we get

$$\int_M \omega^m = \int_M d(\theta \wedge \omega^{m-1}) = \int_{\partial M} \theta \wedge \omega^{m-1} = 0,$$

where we used Stokes' theorem and the fact that M is compact without boundary. However, since ω is a symplectic form, ω^m is a volume form, and thus $\int_M \omega^m > 0$. This is a contradiction, hence $[\omega] \neq 0$. □

Corollary 2.2. Let M be a compact symplectic manifold. Then $H^2_{dR}(M,\mathbb{R}) \neq 0$.

More generally, we have the following result.

Theorem 2.3. Let M be a compact symplectic manifold of dimension $n = 2m$. Then $H^{2k}_{dR}(M,\mathbb{R}) \neq 0$ for all $0 \leq k \leq m$.

Proof. The case $k = 0$ is trivial. We show that $[\omega^k] \neq 0 \in H^{2k}_{dR}(M,\mathbb{R})$ for all $0 < k \leq m$. Suppose the contrary that $[\omega^k] = 0$ for some k. Then there exists a $(2k-1)$-form θ such that $\omega^k = d\theta$. Similar to the previous proof, we have

$$\int_M \omega^m = \int_M \omega^k \wedge \omega^{m-k} = \int_M d\theta \wedge \omega^{m-k} = \int_M d(\theta \wedge \omega^{m-k}) = \int_{\partial M} \theta \wedge \omega^{m-k} = 0,$$

which is a contradiction. □

Example 2.4. The n-dimensional sphere S^n admits a symplectic form if and only if $n = 2$. In fact, if $n = 2$, then S^2 is symplectic with the area form as the symplectic form. If $n \neq 2$, then $H^2_{dR}(S^n,\mathbb{R}) = 0$, so S^n cannot admit a symplectic form.

Example 2.5. The manifold $S^2 \times S^6$ does not admit a symplectic form. This is because $H^4_{dR}(S^2 \times S^6,\mathbb{R}) = 0$, so it has no symplectic form.

Example 2.6. The manifold $S^1 \times S^3$ does not admit a symplectic form since $H^2_{dR}(S^1 \times S^3,\mathbb{R}) = 0$. It is interesting to note that $S^1 \times S^3$ admits a complex structure. Indeed, we have

$$S^3 \times S^1 \cong \left(\mathbb{C}^2 \setminus \{0\}\right)/\Gamma,$$

where Γ is the group generated by the holomorphic map $(z_1, z_2) \mapsto (2z_1, 2z_2)$.

Theorem 2.7. All orientable surfaces admit symplectic forms.

Proof. Let M be an orientable surface. Then M admits a Riemannian metric g. Let ω be the area form of g. Then ω is a symplectic form on M. □

Example 2.8. The vector space $\mathbb{R}^{2n}$ admits a symplectic form given by

$$\omega = dx_1 \wedge dy_1 + \cdots + dx_n \wedge dy_n,$$

where $(x_1, y_1, \ldots, x_n, y_n)$ are the standard coordinates on $\mathbb{R}^{2n}$. This is called the *standard symplectic form* on $\mathbb{R}^{2n}$.

Recall that $\mathbb{C}^n = \mathbb{R}^{2n}$ has a standard complex structure J and a standard Riemannian metric g. One can check that the standard symplectic form ω on $\mathbb{C}^n$ is given by $\omega(X,Y) = g(JX,Y)$ for all $X, Y \in T_z\mathbb{C}^n$. Hence, we say that $(\mathbb{C}^n, \omega, J, g)$ is a Kähler manifold.

Example 2.9. Let $M = \mathbb{P}^n(\mathbb{C})$ be the complex projective space. Recall that we have many descriptions of $\mathbb{P}^n(\mathbb{C})$:

(a) $\mathbb{P}^n(\mathbb{C}) = \{\text{complex lines in } \mathbb{C}^{n+1}\}$;

(b) $\mathbb{P}^n(\mathbb{C}) = \left(\mathbb{C}^{n+1} \setminus \{0\}\right)/\mathbb{C}^*$;

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(c) $\mathbb{P}^n(\mathbb{C}) = S^{2n+1}/S^1$.

Recall that the *Fubini-Study* metric on $\mathbb{P}^n(\mathbb{C})$ is given by the descending of the standard metric on $S^{2n+1} \subset \mathbb{C}^{n+1}$ since the S^1-action on S^{2n+1} is isometric. More explicitly, let $z \in S^{2n+1}$ with image $[z] \in \mathbb{P}^n(\mathbb{C})$. Then the tangent space at $[z]$ is given by

$$T_{[z]}\mathbb{P}^n(\mathbb{C}) = \{X \in \mathbb{C}^{n+1} : \langle X, z\rangle = \langle X, \sqrt{-1}z\rangle = 0\},$$

where $\langle\cdot,\cdot\rangle$ is the standard Euclidean inner product on $\mathbb{C}^{n+1} \cong \mathbb{R}^{2n+2}$. The Fubini-Study metric g on $\mathbb{P}^n(\mathbb{C})$ is given by $g(X,Y) = \langle X, Y\rangle$ for all $X, Y \in T_{[z]}\mathbb{P}^n(\mathbb{C}) \subset \mathbb{C}^{n+1}$. And the almost complex structure J on $\mathbb{P}^n(\mathbb{C})$ is also given by $X \mapsto \sqrt{-1}X$ for all $X \in T_{[z]}\mathbb{P}^n(\mathbb{C}) \subset \mathbb{C}^{n+1}$.

For every $X, Y \in T_{[z]}\mathbb{P}^n(\mathbb{C})$, we define the Fubini-Study symplectic form ω on $\mathbb{P}^n(\mathbb{C})$ by

$$\omega(X,Y) = g(JX,Y) = \langle\sqrt{-1}X, Y\rangle.$$

This is a non-degenerate 2-form on $\mathbb{P}^n(\mathbb{C})$. By direct computation on local charts, we can check that ω is closed, hence it is a symplectic form on $\mathbb{P}^n(\mathbb{C})$. In fact, $(\mathbb{P}^n(\mathbb{C}), \omega, J, g)$ is a Kähler manifold.

Definition 2.10. Let M be a smooth manifold with a(n almost) complex structure J, a Riemannian metric g. We say that (M, J, g) is an *(almost) Kähler manifold* if the following conditions hold:

(a) $g(JX, JY) = g(X,Y)$ for all $X, Y \in TM$;

(b) $\omega(X,Y) := g(JX,Y)$ is a closed 2-form on M.

The 2-form ω is called the *(almost) Kähler form* of (M, J, g). It is easy to check that ω is a symplectic form on M.

Assume that (M, J, g) is a(n almost) Kähler manifold. Let N is a(n almost) complex submanifold of M. Here we say that N is a(n almost) complex submanifold of M if $J(TN) \subset TN$. Then the restriction of ω to N is a(n almost) Kähler form on N. Note that the submanifold N of a symplectic manifold (M, ω) is not necessarily a symplectic submanifold of M since the restriction of ω to N may be degenerate.

Example 2.11. All complex submanifolds of $\mathbb{C}^n$ and $\mathbb{P}^n(\mathbb{C})$ are Kähler manifolds and hence symplectic manifolds.

Example 2.12. Let M be an arbitrary smooth manifold of dimension m. Then the cotangent bundle $T^*M =: N$ admits a natural symplectic form ω. Let $\pi : N \to M$ be the projection map. For every $p \in M$ and $\varphi \in T_p^*M$, and for $v \in T_\varphi N$, $\pi_*(v) \in T_pM$ is well-defined. We define

$$\theta(v \in T_\varphi N) = \varphi(\pi_*(v)) \in \mathbb{R}.$$

Then θ is a 1-form on N, and we define $\omega = d\theta$. Easily see that ω is closed.

For the non-degeneracy of ω, we can work in local coordinates. Let $(U, x_1, \dots, x_m)$ be a local coordinate chart on M. For every $\varphi \in T*U \subset T*M$, we can write $\varphi = \sum_{i=1}^m y_i dx_i$ for some unique tuple $y_1, \dots, y_m \in \mathbb{R}$. Then we have a local coordinate chart $(\pi^{-1}(U), x_1, \dots, x_m, y_1, \dots, y_m)$ on N. In this local coordinate chart, we have $\theta = \sum_{i=1}^m y_i dx_i \in T^*N$. Hence, we have

$$\omega = d\theta = \sum_{i=1}^m dy_i \wedge dx_i.$$

This is a non-degenerate 2-form on N.

Example 2.13. If M_i, ω_i are symplectic manifolds for $i = 1, 2$, then the product manifold $M_1 \times M_2$ admits a natural symplectic form $\omega = \pi_1^* \omega_1 + \pi_2^* \omega_2$, where $\pi_i : M_1 \times M_2 \to M_i$ is the projection map for $i = 1, 2$.

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